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# Advanced Textbooks in Control and Signal Processing

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Bruno Siciliano · Luigi Villani  
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# Foundations of Robotics

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*To our students*

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## Series Editor's Foreword

Robotics stands out among engineering fields for its uniqueness, acting as a cross-roads for various scientific and technical disciplines. The difficulty in defining a precise boundary for robotics arises from its inherently multidisciplinary nature: it does not belong to a single field but integrates and harmonizes contributions from a wide range of sectors. Originally closely tied to mechanical engineering, which provided the foundation for the design and construction of the first robots, robotics has rapidly evolved, incorporating essential elements from control engineering and electronics. Today, robotics is heavily influenced by computer science and computational sciences. The ability to process and interpret data, execute advanced algorithms, and integrate artificial intelligence has revolutionized how robots interact with their environment and with humans, significantly expanding the engineering and social domains in which robots are employed.

The series *Advanced Textbooks in Control and Signal Processing* warmly embraces these disciplinary areas, where various technological and scientific domains intersect, each with its own characteristics, impacting numerous productive and social sectors.

The variety of scenarios and the richness of the disciplines involved present the challenge of addressing the topic systematically and orderly, avoiding the pitfall of merely listing tools and solutions in a disorganized manner. The textbook by authors Bruno Siciliano, Luigi Villani, Giuseppe Oriolo, and Alessandro De Luca masterfully addresses this issue by focusing on the three crucial aspects underlying the development of robotic systems and, more broadly, of any engineering system: modeling, planning, and control. These three areas are treated rigorously, demonstrating how different branches of mathematics and engineering contribute with various tools. The discussion is never 'sequential' but pays particular attention to showing how choices in these three areas are often interconnected, promoting integrated and 'circular' design. This structure allows the authors to explore a wide range of presented tools while maintaining the systematic approach essential for integrated design.

This new book can be seen as the significant extension of *Robotics: Modelling, Planning and Control* by Bruno Siciliano, Luigi Villani, and Giuseppe Oriolo with Lorenzo Sciacivco (2009). Compared with this, the new book reorganizes and

significantly improves the topics already covered previously and adds two new relevant chapters dealing with flexible-joint robots and cooperative manipulation, consistently addressed in terms of ‘modeling, planning, and control’. This new book undoubtedly establishes itself as an international reference in both educational and research contexts and as an undisputed flagship for the series *Advanced Textbooks in Control and Signal Processing*.

Bologna, Italy  
November 2024

Lorenzo Marconi

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## Preface

After more than 60 years of extraordinary progress in research and applications, Robotics is now considered a science in all respects, with its principles, theories, and methods. Robotics roots are from several cultural domains, such as cybernetics, mechatronics, computer science, and bioengineering. This impressive multidisciplinary nature will be a source of inspiration also for the next generations of autonomous robots—and for roboticists who will develop them. To guarantee continued future success, fundamental scientific material should be made available in a clear, technically precise, and complete fashion.

Based on the educational and research experience of the authors, this book presents the *Foundations of Robotics* as concerns modeling, planning, and control of robot manipulators and mobile robots, whose domain encompasses industrial as well as advanced service and field applications. The book is a natural evolution of the previous textbook *Robotics: Modelling, Planning and Control* by Bruno Siciliano, Luigi Villani, and Giuseppe Oriolo, together with Lorenzo Sciavicco, published in 2009. Further to fully revising, reorganizing, and substantially extending the contents, new topics have been added such as joint elasticity and cooperative manipulation.

The book is organized in 12 chapters. Each chapter has a brief preamble that provides the rationale and objectives of the subject matter.

Chapter 1 introduces the subject with a brief history of robotics from its industrial uses in manufacturing to advanced service and field applications. The mechanical structures of manipulators, wheeled vehicles, and other robotic systems are illustrated, and the main actuation and sensing units are described, together with the general control architectures.

Chapter 2 gives the basic tools for representing the position and orientation of rigid bodies. The notions of configuration space and task space are introduced: the geometric relationship between these two spaces provides the direct kinematics of a robot. For manipulators, this is derived within the classical Denavit–Hartenberg convention—including an extension to closed kinematic chains. The issues of kinematic calibration, robot redundancy with respect to a task, and reachable workspace are also discussed. The inverse kinematics problem is analyzed, and closed-form solutions are found for typical manipulation structures.



Differential kinematics is presented in Chap. 3. For a manipulator, the relationship between joint velocities and end-effector velocities (twists) is described by the geometric Jacobian. In general, the concept of analytic Jacobian associated with a task is introduced. These mathematical tools characterize the mobility of a robot, with the determination of singular configurations, the definition of manipulability indices, and with the dual relationship in static conditions between generalized forces in the task space and the corresponding joint torques. The inverse differential kinematics problem is addressed in more detail for redundant robots, where optimization-based solutions and task augmentation or prioritization methods are presented. The Jacobian matrix is also at the core of numerical algorithms that solve the inverse kinematics problem, e.g., in robots not having a closed-form solution.

Trajectory planning techniques for robot manipulators are illustrated in Chap. 4. The problem is decomposed into two parts: planning a parametrized geometric path, typically by low-order interpolating polynomials through a sequence of via points with boundary conditions; and defining a suitable timing law for tracing the path, possibly followed by uniform time scaling in the presence of motion limits. Point-to-point and multipoint problems are considered, first in the configuration space and then in the task space, where special concern is devoted to the end-effector orientation.

Chapter 5 deals with the derivation of the robot dynamic model, which plays a key role in the design and analysis of a manipulation structure, in performing numerical simulations and in the synthesis of motion control algorithms. The direct and inverse dynamics problems are first formulated. Two modeling approaches are considered, the energy-based Euler–Lagrange formulation, which lends itself to a symbolic derivation, and the Newton–Euler formulation based on the balance of forces and moments, which can be computed efficiently by a recursive numerical scheme. Notable properties of the dynamic model are presented, including dynamic scaling of trajectories and linearity in the parameters, which is used to develop the identification of the essential dynamic coefficients of the robot. The two formulations are compared in terms of their use for direct and inverse dynamics problems. Finally, robot dynamics is also derived in the task space and in the presence of additional geometric constraints (by projection or reduction methods).

Chapter 6 discusses the control problem for manipulators moving in free space. A single-joint case is treated first to introduce the relevant feedback and feedforward actions used in the independent joint control technique, which is typically found in industrial robots. The regulation problem, i.e., stabilizing the robot to a desired equilibrium configuration, is tackled by PD control with gravity compensation or cancellation, or PID control using energy arguments and Lyapunov analysis. Trajectory tracking problems are then addressed by different methods that fully consider the nonlinear dynamics of the robotic system, including feedback linearization, inverse dynamics control, adaptive control, and robust control. Selected methods are extended to the motion control problem defined in the task space, with extra considerations for the case of redundant robots.

Wheeled mobile robots are dealt with in Chap. 7, which extends some modeling, planning, and control aspects of previous chapters. The kinematic model is derived from the nonholonomic constraints imposed by the pure rolling of the robot wheels. The methods for planning admissible paths exploit the model structure, using suitable transformations and input parametrizations. The dynamic model of wheeled mobile robots is derived as well. Motion control laws are presented for trajectory tracking and for the harder problem of regulation to a configuration. The localization of mobile robots is addressed using odometry or by means of an extended Kalman filter. Modeling and kinematic control of manipulators mounted on wheeled vehicles is also discussed at the end.

In Chap. 8, visual feedback control is introduced which uses some geometric and qualitative information about the environment acquired online by a camera to control the robot motion. After presenting basic algorithms of image processing that extract useful feature parameters from the image, the problem of camera pose estimation with respect to a target is addressed using analytic and numeric techniques. A conceptual solution to the camera calibration problem is also presented. Then, two main approaches to visual robot control are considered, namely position-based and image-based visual servoing. Additional information can be obtained using stereo vision techniques and epipolar geometry, leading to hybrid visual servoing methods that combine the benefits of image-based and position-based approaches.

Chapter 9 reconsiders the path and trajectory planning problems treated in Chaps. 4 and 7, respectively, for robot manipulators and mobile robots, when obstacles are present in the workspace and the robot should execute the assigned task without collision. For this motion planning problem, which is best formulated in the configuration space, several planning algorithms are presented for the case of mobile robots, including exact (retraction, cell decomposition), probabilistic (PRM, RRT), and heuristic (artificial potential) methods. Extension to the case of robot manipulators is also discussed.

The physical interaction between a manipulator and its environment is considered in Chap. 10, where the contact force is the additional quantity to be set under control. For this, the concepts of mechanical admittance and impedance are used, and the stability of the coupled robot–environment system is analyzed by passivity arguments. Suitable control strategies are aimed at reshaping in a desired way the robot–environment interaction dynamics. When the environment imposes geometric constraints, the interaction task is defined by natural and artificial (control) constraints. Hybrid force/motion control schemes are presented, where motion and force are controlled along different task directions. Finally, such approach is extended from rigid to compliant environments.

In Chap. 11 the assumption of rigid transmissions between the driving motors and the driven rigid links is relaxed. The dynamic modeling of manipulators with elastic joints is presented, addressing the peculiarities of the inverse dynamics

problem in this case. Regulation and trajectory tracking laws must deal with the underactuated nature of such mechanical systems. Lyapunov and feedback linearization control methods are suitably extended to this class of robots.

Chapter 12 covers cooperative manipulation, namely the kinematics, statics, dynamics, and control of two or more robots manipulating a common object. The grasp matrix along with the internal and external object wrenches are central concepts in this case. Control strategies are concerned with cooperative motion and interaction forces between manipulators and the grasped object. The theory is then applied to dexterous object manipulation with multifingered hands, where contact and friction models specify the set of transmitted twists and contact wrenches. A hand Jacobian is defined for this purpose, and the related grasp properties allow for a classification that is useful for manipulation planning.

In conclusion, five appendices review the basic methodologies used in the book to provide readers of different extraction with a homogeneous background.

Appendix A is devoted to linear algebra and presents the fundamental notions on vectors and matrices, and the related properties and operations. The basic concepts of rigid-body mechanics, which are preliminary to the study of manipulator dynamics, are recalled in Appendix B. Tools from differential geometry used in the control of mechanical systems subject to nonholonomic constraints are given in Appendix C. Appendix D illustrates the principles of feedback control of linear and nonlinear systems, including Lyapunov stability analysis, passivity properties of interconnected systems, exact linearization by nonlinear feedback, and use of Kalman filtering for state estimation. Finally, Appendix E focuses on graph search algorithms and complexity in view of their application to motion planning methods.

The organization of the contents allows the adoption of the book as a reference text for senior undergraduate or graduate courses in automation, computer science, and electrical, electronics, or mechanical engineering with strong robotics content. A typical senior undergraduate course may consist of Chaps. 1–4 (including kinematic control), possibly with some extensions concerning dynamics (from Chap. 5) and independent joint control (Sects. 6.2 and 6.3.1). A graduate course may start from Chap. 5 on, in the case with a preferred final choice among Chaps. 9, 11 or 12. A course on mobile robotics may cover Chaps. 7–9, with some selected materials extracted from Chaps. 1 and 3.

The book contains 245 figures, 14 tables, and 52 worked-out examples and case studies spread throughout the text with the use of numerical simulations. In addition, 193 end-of-chapter problems are proposed, some of which contain further study material of the contents, and the book is accompanied by an electronic solutions manual; this is available free of charge to those adopting this volume as a text for courses (downloadable from <https://sites.google.com/springernature.com/extramaterial/lecturer-material>). Special care has been devoted to the

selection of bibliographical references (290), which are cited at the end of each chapter in relation to the historical development of the field.

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## Abbreviations and Symbols

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### Abbreviations

|      |                                                |
|------|------------------------------------------------|
| 2D   | Two-Dimensional                                |
| 3D   | Three-Dimensional                              |
| A/D  | Analog-to-Digital                              |
| AAR  | Angular Acceleration Recursion                 |
| AC   | Alternate Current                              |
| AGV  | Automatic Guided Vehicle                       |
| AI   | Artificial Intelligence                        |
| AR   | Acceleration Recursion                         |
| AVR  | Angular Velocity Recursion                     |
| BDF  | Backward Differentiation Formula               |
| BFS  | Breadth-First Search                           |
| CCD  | Charge Coupled Device                          |
| CCIR | Consultative Committee for International Radio |
| CLIK | Closed-Loop Inverse Kinematics                 |
| CMA  | Center-of-Mass Acceleration                    |
| CMOS | Complementary Metal Oxide Semiconductor        |
| CoM  | Center of Mass                                 |
| DC   | Direct Current                                 |
| DFS  | Depth-First Search                             |
| DH   | Denavit–Hartenberg                             |
| DLS  | Damped Least Squares                           |
| DoF  | Degree of Freedom                              |
| EKF  | Extended Kalman Filter                         |
| EL   | Euler–Lagrange                                 |
| FIFO | First In First Out                             |
| FMS  | Flexible Manufacturing System                  |
| fps  | Frames per second                              |
| FR   | Force Recursion                                |
| GF   | Generalized Force                              |
| GPS  | Global Positioning System                      |

|                  |                                         |
|------------------|-----------------------------------------|
| HF               | Hard Finger                             |
| IBVS             | Image-Based Visual Servoing             |
| IMU              | Inertial Measurement Unit               |
| ITS              | Intelligent Transportation System       |
| KF               | Kalman Filter                           |
| KKF              | Kinematic Kalman Filter                 |
| LIDAR            | Laser Imaging Detection And Ranging     |
| LIFO             | Last In First Out                       |
| LQ               | Linear–Quadratic                        |
| MP               | Multi-Point                             |
| MR               | Moment Recursion                        |
| NE               | Newton–Euler                            |
| NEA              | NE Algorithm                            |
| PBVS             | Position-Based Visual Servoing          |
| PD               | Proportional–Derivative                 |
| PI               | Proportional–Integral                   |
| PID              | Proportional–Integral–Derivative        |
| P <sub>n</sub> P | Perspective- <i>n</i> -Point            |
| PR               | Positive Real                           |
| PRM              | Probabilistic RoadMap                   |
| PTP              | Point-to-Point                          |
| PUMA             | Programmable Universal Manipulation Arm |
| PwoF             | Point without Friction                  |
| RCC              | Remote Center of Compliance             |
| RFID             | Radio Frequency Identification          |
| RGB              | Red–Green–Blue                          |
| RGB-D            | RGB–Depth                               |
| rpm              | Revolutions per minute                  |
| RPY              | Roll–Pitch–Yaw                          |
| RRT              | Rapidly-exploring Random Tree           |
| SCARA            | Selective Compliance Assembly Robot Arm |
| SEA              | Series Elastic Actuation                |
| SF               | Soft Finger                             |
| SI               | Système International                   |
| SPR              | Strictly Positive Real                  |
| SVD              | Singular Value Decomposition            |
| TCP              | Tool Center Point                       |
| UAV              | Unmanned Aerial Vehicle                 |
| UUV              | Unmanned Underwater Vehicle             |
| VSA              | Variable Stiffness Actuation            |
| ZOH              | Zero-Order Hold                         |

## Symbols

|              |                                                                      |
|--------------|----------------------------------------------------------------------|
| $\mathbf{0}$ | Zero vector                                                          |
| $\alpha$     | DH parameter                                                         |
| $\beta$      | Friction vector in task space                                        |
| $\gamma$     | Configuration clearance                                              |
| $\gamma_0$   | Obstacle range of influence                                          |
| $\gamma$     | Vector of contact forces and moments                                 |
| $\Gamma$     | Factorization matrix of Coriolis and centrifugal terms in task space |
| $\Delta$     | Distribution                                                         |
| $\epsilon$   | Vector part of unit quaternion                                       |
| $\varsigma$  | Damping factor                                                       |
| $\eta$       | Path torsion                                                         |
|              | Scalar part of unit quaternion                                       |
| $\eta$       | Gravity vector in task space                                         |
|              | Static friction coefficient                                          |
| $\mu$        | Vector of moments                                                    |
| $\vartheta$  | Rotation angle in axis–angle representation                          |
| $\theta$     | DH parameter                                                         |
| $\theta_m$   | Motor angle                                                          |
| $\theta$     | Vector of motor positions                                            |
| $\kappa$     | Path curvature                                                       |
| $\lambda$    | Matrix eigenvalue                                                    |
|              | Scaling factor in perspective transformation                         |
| $\lambda$    | Vector of Lagrange multipliers                                       |
| $\Lambda$    | Inertia matrix in task space                                         |
| $\nabla$     | Nabla                                                                |
| $\pi$        | Vector of standard dynamic parameters                                |
| $\Pi$        | Matrix in perspective transformation                                 |
| $\varrho$    | Vector of dependent joint variables in closed kinematic chains       |
| $\rho$       | Vector of dynamic coefficients                                       |
| $\xi$        | Vector of nonconservative generalized forces                         |
| $\sigma$     | Arc length                                                           |
|              | Matrix singular value                                                |
| $\Sigma$     | Matrix in SVD                                                        |
| $\tau$       | Vector of joint torques                                              |
| $\Upsilon$   | Matrix for expressing joint velocities in closed kinematic chains    |
| $\phi$       | Vector of Euler angles                                               |
| $\Phi$       | Regressor matrix in calibration                                      |
| $\psi$       | Coriolis and centrifugal vector in task space                        |
| $\omega$     | Angular velocity vector                                              |
| $\Omega$     | Matrix of camera intrinsic parameters                                |
| $\omega_n$   | Natural frequency                                                    |
| $a$          | DH parameter                                                         |

---

|                             |                                                          |
|-----------------------------|----------------------------------------------------------|
| $\mathbf{a}$                | Acceleration vector                                      |
|                             | Unit approach vector                                     |
| $\mathbf{A}$                | DH homogeneous transformation matrix                     |
|                             | Matrix of Pfaffian constraints                           |
| arg                         | Argument of a function                                   |
| atan2                       | Four-quadrant inverse tangent                            |
| $\mathbf{b}$                | Unit binormal vector                                     |
| $\mathbf{B}$                | Viscous friction matrix                                  |
| $\mathcal{B}$               | Robot body                                               |
| $c_{ijk}$                   | Christoffel symbol                                       |
| $\mathcal{C}$               | Connectivity graph                                       |
| $\mathbf{c}$                | Vector of Coriolis and centrifugal torques               |
| $\mathbf{C}$                | Compliance matrix                                        |
|                             | Factorization matrix of Coriolis and centrifugal torques |
| $\mathcal{C}$               | Configuration space                                      |
| $\mathcal{C}_{\text{free}}$ | Free configuration space                                 |
| card                        | Cardinality of set                                       |
| $\mathcal{CO}$              | $\mathcal{C}$ -obstacle region                           |
| $d$                         | DH parameter                                             |
| $\mathbf{D}$                | Damping matrix in elastic joints                         |
| det                         | Determinant of matrix                                    |
| diag                        | Diagonal matrix                                          |
| dim                         | Dimension of space                                       |
| $\mathbf{e}$                | Error vector                                             |
| $\mathbf{E}$                | Essential matrix in stereo vision                        |
| $f$                         | Focal length                                             |
| $\mathbf{f}$                | Force vector                                             |
| $g$                         | Gravity acceleration                                     |
| $\mathbf{g}$                | Vector of gravitational torques                          |
| $\mathbf{g}_0$              | Gravity acceleration vector                              |
| $\mathbf{G}$                | Grasp matrix                                             |
| $\mathbf{H}$                | Matrix for finger contact model                          |
|                             | Planar homography matrix                                 |
| $\mathcal{H}$               | Hamiltonian                                              |
| $i$                         | Current                                                  |
| $\mathbf{I}$                | Identity matrix                                          |
|                             | Inertia tensor                                           |
| Int                         | Interior of set                                          |
| $\mathbf{J}$                | Analytical Jacobian                                      |
| $\mathbf{J}_e$              | End-effector Jacobian                                    |
| $\mathbf{J}_f$              | Complete hand Jacobian                                   |
| $\mathbf{J}_g$              | Geometric Jacobian                                       |
| $\mathbf{J}_h$              | Hand Jacobian                                            |
| $\mathbf{J}_s$              | Image Jacobian                                           |
| $\mathbf{k}$                | Direct kinematics function                               |



---

|                |                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{K}$   | Stiffness matrix                                                                                                               |
| $\ell_C$       | Distance between link CoM and previous joint axis                                                                              |
| $L$            | Lie derivative                                                                                                                 |
| $\mathbf{L}$   | Interaction matrix in visual servoing                                                                                          |
| $\mathcal{L}$  | Lagrangian                                                                                                                     |
| $m$            | Mass                                                                                                                           |
| $\mathbf{M}$   | Inertia matrix                                                                                                                 |
| $\mathcal{M}$  | Inertia matrix in robots with elastic joints                                                                                   |
| max            | Maximum                                                                                                                        |
| min            | Minimum                                                                                                                        |
| $n_r$          | Gear reduction ratio                                                                                                           |
| $\mathbf{n}$   | Unit normal vector                                                                                                             |
| $\mathcal{N}$  | Null space of matrix                                                                                                           |
| $\mathcal{O}$  | Obstacle in workspace                                                                                                          |
| $\mathbf{O}$   | Zero matrix                                                                                                                    |
| $\mathbf{p}$   | Generalized momentum vector                                                                                                    |
|                | Position vector                                                                                                                |
| $\mathbf{P}$   | Orthogonal projection matrix                                                                                                   |
| $\mathcal{P}$  | Power                                                                                                                          |
| $\mathbf{q}$   | Configuration or joint vector                                                                                                  |
| $\mathbf{q}_e$ | Configuration vector for manipulators with elastic joints                                                                      |
| $\mathcal{Q}$  | Unit quaternion                                                                                                                |
| $\mathbf{r}$   | Position vector between origins of two frames/unit vector in axis–angle representation                                         |
| $\mathbf{R}$   | Rotation matrix                                                                                                                |
| $\mathcal{R}$  | Range space of matrix/roadmap in configuration space                                                                           |
| rank           | Rank of matrix                                                                                                                 |
| $\mathbb{R}^n$ | Euclidean space of dimension $n$                                                                                               |
| $s$            | Path parameter                                                                                                                 |
| $\mathbf{s}$   | Image feature vector/unit sliding vector                                                                                       |
| $\mathbf{S}$   | Skew-symmetric matrix                                                                                                          |
| $SE(m)$        | Special Euclidean group of rigid motions in $\mathbb{R}^m$                                                                     |
| <b>sign</b>    | Vector sign function                                                                                                           |
| $SO(m)$        | Special group of $m \times m$ orthonormal matrices                                                                             |
| span           | Linear span of set of vectors                                                                                                  |
| $t$            | Time                                                                                                                           |
| $\mathbf{t}$   | Twist vector                                                                                                                   |
| $T$            | Time interval                                                                                                                  |
|                | Tree                                                                                                                           |
|                | Unit tangent vector                                                                                                            |
| $\mathbf{T}$   | Homogeneous transformation matrix                                                                                              |
| $\mathbf{T}_E$ | Transformation matrix from Euler angle rates to angular velocity/<br>transformation matrix from geometric to analytic Jacobian |
| $\mathcal{T}$  | Kinetic energy                                                                                                                 |
|                | Task space                                                                                                                     |

|                  |                                                  |
|------------------|--------------------------------------------------|
| trace            | Trace of matrix                                  |
| $U$              | Artificial potential field                       |
| $\mathcal{U}$    | Potential energy                                 |
| $\mathcal{U}_e$  | Elastic potential energy                         |
| $\mathcal{U}_g$  | Gravitational potential energy                   |
| $v$              | Voltage                                          |
| $\mathbf{v}$     | Velocity vector                                  |
| $V$              | Lyapunov function                                |
| $\mathcal{V}$    | Generalized Voronoi diagram                      |
| $\mathbf{w}$     | Wrench vector                                    |
| $W$              | Work                                             |
| $\mathcal{W}$    | Workspace                                        |
| $\mathcal{W}_r$  | Reachable workspace                              |
| $\mathbf{y}$     | Task vector                                      |
| $Y$              | Admittance                                       |
| $\mathbf{Y}$     | Regressor matrix for dynamic coefficients        |
| $\mathbf{Y}_\pi$ | Regressor matrix for standard dynamic parameters |
| $Z$              | Impedance                                        |